

Fault Diagnosis Using Improved Discrimination Locality Preserving Projections Integrated With Sparse Autoencoder

Yan-Lin He(贺彦林)^{ID}, *Member, IEEE*, Kun Li(李琨)^{ID}, Ning Zhang(张宁)^{ID}, Yuan Xu(徐圆)^{ID},
and Qun-Xiong Zhu(朱群雄)^{ID}

Abstract—In order to ensure safe operations of industrial processes, it is especially important to find fault types accurately and quickly based on historical data and then deal with faults in time. Unfortunately, due to the tricky characteristics of industrial process data such as massive number, high dimensionality, and nonlinearity, it turns out that timely and accurate diagnosis of faults becomes of great difficulty in industrial processes. To address this problem, novel effective fault diagnosis using an improved global and local dimensionality reduction (DR) method named discrimination locality preserving projections integrated with sparse autoencoder (SAEDLPP) is proposed in this article. In SAEDLPP, the global DR information of data is first obtained by sparse autoencoder (SAE); next, the DR data obtained through SAE are passed through discrimination locality preserving projections (DLPP) to obtain local DR information, preserving not only the global information but the local information of the extracted features. Finally, fault diagnosis is achieved by separating the extracted features by SAEDLPP using an AdaBoost classifier to recognize fault types. Simulations are conducted on the Tennessee Eastman process (TEP) and the results indicate that the provided SAEDLPP-based fault diagnosis methodology can achieve much higher accuracy in fault diagnosis than other associated methods.

Index Terms—Dimensionality reduction (DR), discriminant locality preserving projections (LPPs), fault diagnosis, industrial processes, sparse autoencoder (SAE).

I. INTRODUCTION

WITH the development of science and technology and the progress of society, industrial processes are gradually becoming more and more large-scale and complex. If the production process breaks down and faults are not handled in time, huge damage may occur. Therefore, it is particularly important to utilize effective methods of judgment and troubleshooting for timely and reliable diagnosis of process faults [1]. Currently, for the industrial process, fault diagnosis technology has become a popular research direction. Since modern industrial process data are massive, high dimensional,

and nonlinear, it is difficult to analyze process data directly. Thus, it is especially important to reduce the dimensionality of process data for extracting useful features in the first place [2]. With the fast development of sensor and computer technologies, industrial process data can be easily collected and well stored in real time. Therefore, data-driven techniques of fault diagnosis based on historical process data have been broadly developed and utilized [3].

Common dimensionality reduction (DR) methods like principal component analysis (PCA) [4], fisher discriminant analysis (FDA) [5], partial least squares (PLSs) [6], and independent component analysis (ICA) [7] have been successfully and widely used in fault detection and fault diagnosis of complex processes. A method of fault detection and diagnosis based on PCA was proposed by Li *et al.* [8]. PCA does not consider the correlation between faults. For handling this problem, Yin *et al.* [6] proposed a process monitoring approach named PLS. PCA and PLS do not consider the discriminative information between classes. What is more, the performance of PCA and ICA relies on the distribution characteristics of data. FDA solves the problem well by adding the discriminative information to its objective function [9]. However, the data in industrial processes are time-series and dynamic in nature. Thus, these static methods do not solve the problem of dynamic nature. Therefore, methods based on dynamic PCA [10], dynamic FDA [11], dynamic PLS [12], etc., were proposed to solve the problem of dynamic data. Unfortunately, the above methods do not take into account the manifold structure between data, resulting in fault diagnosis with low accuracy.

The above traditional methods only consider the global Euclidean structure, and so it is hard to discover the low-dimensional manifold embedding existing in the high-dimensional data. That is to say, the above traditional methods cannot fully extract useful information from high-dimensional data. For addressing this issue, manifold learning-based techniques and methods have been broadly studied and applied [13], [14]. Common manifold learning techniques like isometric mapping (ISOMAP) [15], locally linear embedding (LLE) [16], and locality preserving projections (LPPs) [17] have been successfully utilized. The manifold learning techniques can discover the manifold structure from high-dimensional data with low-dimensional space when performing DR. One of the most commonly used manifold

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The authors are with the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing 100029, China (e-mail: hey1@mail.buct.edu.cn; 1164609251@qq.com; zhangnieng@126.com; xuyuanbuct@163.com; zhuqx@mail.buct.edu.cn).

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learning models is the LPP algorithm. LPP can find the streaming embedding from high-dimensional samples in low-dimensional space very well; however, the LPP method ignores the discriminative information between classes, limiting its performance in classification problems. To improve the classification performance of LPP, Yu *et al.* [18] introduced the idea of FDA into LPP and then proposed a method called discrimination LPPs (DLPP). Although DLPP has been successfully applied in kinds of fields, DLPP encounters the problem of matrix decomposition [19]. To solve the problem of matrix decomposition, a method that combined PCA with DLPP was proposed [20]. In the PCA + DLPP method, useful information may be lost in the PCA stage; in addition, PCA is a linear model which cannot handle the nonlinearity of process data. When the fault class number is less than the data dimension, PCA + DLPP achieves unsatisfactory classification accuracy.

As linear tools, manifold learning techniques are generally based on local operations, leading to poor DR when facing massive high-dimensional data. However, the manifold learning algorithms based on global operations pay a fat price in terms of computational amount and computational difficulty. Therefore, the concept of nonlinear DR should be considered. For nonlinear DR, neural networks can be adopted as effective methods of nonlinear global DR. Common neural network DR and feature extraction methods include convolutional neural networks (CNNs) [21], sparse autoencoder (SAE) [22], and self-organizing mapping (SOM) [23]. CNNs are commonly used in image processing; autoencoder-based nonlinear dimension reduction methods have been applied in a number of areas. Common autoencoders include denoising autoencoders [24] and stack autoencoders [25]. Although neural network-based nonlinear DR methods have many good features, the neural network-based nonlinear DR methods do not take into account the manifold structure of data. Traditional manifold learning-based DR algorithms may achieve good DR for certain problems. However, some traditional manifold learning-based DR algorithms are linear and so cannot well deal with highly nonlinear data.

In order to get good feature extraction performance, it has become a new trend to reasonably combine multiple DR methods for DR and feature extraction. The performance of the PCA + DLPP proposed in [20] is still much limited: PCA and DLPP are linear models which cannot handle the nonlinearity of process data well; the problem of matrix decomposition in DLPP is not fully solved. Generalized autoencoder (GAE) is a method that combines a self-encoder with a manifold learning method [26]. Thus, drawing lessons from GAE and based on the above analysis, this article proposes an improved global and local DR method named DLPPs integrated with SAE (SAEDLPP). SAEDLPP integrates SAE with DLPP. The advantages of this kind of integration are that the global DR information of data can be first obtained by extracting features from data through SAE; next, the local DR information of data can be obtained by developing DLPP for the DR data obtained by SAE. In addition, to further solve the problem of matrix decomposition in SAEDLPP, this article decomposes the objective matrix of DLPP using singular value decomposition (SVD). After obtaining the extracted

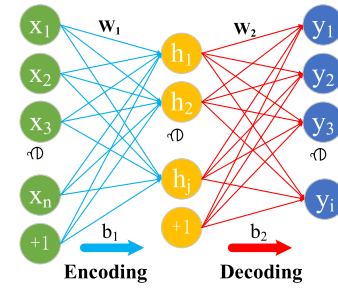


Fig. 1. Structure of autoencoder.

features using the presented SAEDLPP, fault diagnosis can be finally achieved by separating the extracted features through an AdaBoost classifier to recognize fault types. To select the optimal DR order, the Akaike Information Criterion (AIC) is adopted. All in all, a novel and effective model of fault diagnosis is developed based on SAEDLPP and the AdaBoost classifier. To verify and confirm the superior performance of the presented SAEDLPP-based fault diagnosis, the Tennessee Eastman process (TEP) is utilized for experimental analyses. The experimental results show the introduced SAEDLPP-based fault diagnosis can achieve much higher fault diagnosis accuracy compared with other related models, indicating the proposed SAEDLPP-based fault diagnosis model can be utilized as an effective model for recognizing fault types in the field of large-scale industrial processes.

The rest of the article is organized below: a brief introduction to the autoencoder, the basic DLPP method, and the basic classification technique of AdaBoost are presented in Section II; in Section III, the proposed fault diagnosis methodology using the proposed SAEDLPP is described in detail; experiments and result analyses using different fault classes of TEP are provided in Section IV and conclusions are given in Section V.

II. PRELIMINARY METHODS

A. Sparse Autoencoder

SAE [22] is a kind of neural network with one or more hidden layers. SAE adopts an unsupervised learning algorithm. SAE consists of the input layer, the hidden layer, and the output layer, where the input to the hidden indicates the encoding process and the hidden to the output indicates the decoding process. The network structure of SAE is shown in Fig. 1. The idea of SAE is to make sure that the outputs are equal to the inputs, which can be achieved by adjusting the number of neurons in the hidden layer so as to encode different dimensions. The features learned from the autoencoder can preserve important information of original data.

SAE adds an extra penalty factor to the loss function of the traditional autoencoder. SAE can process high-dimensional and nonlinear data and find out the main features in the original input data through the encoding and decoding processes. SAE can discriminate the sparse features of data in the decoding process by adding an extra penalty term so that the parameters of the encoder and decoder can be adjusted to minimize the errors between outputs and inputs [22].

B. Discriminant LPPs

DLPP [19] is a manifold learning DR algorithm based on the popular learning algorithm LPP. The idea of the FDA is added to LPP to develop DLPP. DLPP can reflect the discriminative information between and within samples in the objective function so that the optimal subspace information can be obtained, improving classification ability.

For a given dataset X , suppose a total of n samples are included and the data are divided into K categories. Each class has n_k samples. Then the dataset X can be expressed as

$$X = \{x_1^1, x_2^1, \dots, x_{n_1}^1, x_1^2, x_2^2, \dots, x_{n_2}^2, \dots, x_1^K, x_2^K, \dots, x_{n_K}^K\} \quad (1)$$

where $x_i^k \in \mathbf{R}^m$ means the i th observation of the k th class and each observation is an m -dimensional vector; n_k represents the number of samples in the class k , i.e., $n = n_1 + n_2 + \dots + n_i + \dots + n_K$.

The purpose of DLPP is to find a projection matrix $V \in \mathbf{R}^{m \times a}$ ($a \leq m$) and $y_i = V^T x_i^k, i = 1, 2, \dots, n_k, c = 1, 2, \dots, K$. Details about DLPP can be found in [19].

C. AdaBoost Classification

The AdaBoost classifier is different from the single classifier such as the Bayesian classifier and the decision tree classifier. AdaBoost belongs to an integrated learning classifier. AdaBoost adopts the iteration idea; in each iteration, only one weak classifier is trained and the weak classifier trained in this iteration will participate in the next iteration process. N weak classifiers are trained by N iterations. When the $N + 1$ st iteration is performed, the parameters of the first N iterators do not change and only the data that are not sorted by the first N iterators are classified. Eventually, all trained weak classifiers are linearly combined to make a strong classifier for achieving good classification. Thus, the AdaBoost classifier is adopted. Details about the AdaBoost classifier can be found in [27].

III. PROPOSED SAEDLPP FAULT DIAGNOSIS

In this section, the specific solution of the proposed SAEDLPP is presented. In order to overcome the matrix decomposition problem in DLPP, the feature extraction of the original data with the proposed method SAEDLPP is adopted first and then SVD is used in the matrix decomposition. Finally, the classification is performed by AdaBoost which can be used as a good tool to solve the problem of fault diagnosis.

A. Global Feature Extraction Using SAE

For a given dataset X and the concrete expression in (1), n means the sample number and m means the dimensionality of the samples. The fault data are normalized and the normalization formula is shown in the following equation:

$$x_i = \frac{x_i^k - \bar{x}_i^k}{\sigma} \quad (2)$$

where x_i^k represents each sample, $\bar{x}_i^k$ represents the mean of all samples, and σ represents the sample variance.

The input to the SAE is $X = \{x_1, x_2, \dots, x_i, \dots, x_n\}$ and the output is $Y = \{y_1, y_2, \dots, y_i, \dots, y_n\}$. $x_i, y_i \in \mathbf{R}^m$ are m -dimensional vectors. The loss function for SAE is shown in the following equation:

$$J_{\text{SAE}}(W, b) = \frac{1}{n} \sum_{i=1}^n \frac{1}{2} \|y_i - x_i\|^2 + \frac{\lambda}{2} (\|W\|^2 + \|W'\|^2) + \beta \sum_{j=1}^Z KL(\rho || \hat{\rho}_j) \quad (3)$$

where Z means the number of hidden neural sources in the hidden layer, i denotes the i th input sample, n means the training sample number, λ is the penalty term weight, β is the sparsity weight, and j represents the individual neurons in the hidden layer. The sparse penalty term is shown in the following equation:

$$\beta \sum_{j=1}^{s_2} KL(\rho || \hat{\rho}_j). \quad (4)$$

Adding (4) to the objective function will make the features obtained from learning more binding rather than simply repeating the inputs. The expression for $KL(\rho || \rho_j)$ is given in the following equation:

$$KL(\rho || \rho_j) = \rho \log \frac{\rho}{\rho_j} + (1 - \rho) \log \frac{1 - \rho}{1 - \rho_j} \quad (5)$$

where ρ means the sparsity parameter; the parameter is usually a value close to zero and $\rho_j = \sum_{j=1}^n (1/n) h_j(x_j)$, where ρ_j means the average activation of the hidden layer neurons.

Suppose that the number of nodes in the hidden layer of SAE is Z . Then the hidden layer data $X_{\text{SAE}} = \{x_{\text{SAE}1}, x_{\text{SAE}2}, \dots, x_{\text{SAE}n}\}$ can be obtained after SAE learning, where the expression for c is shown (6). $x_{\text{SAE}i} \in \mathbf{R}^z$ belongs to the z -dimensional vector. In (6), w_1, b_1 are the weights and biases of SAE, respectively

$$x_{\text{SAE}i} = \text{sigmoid}(w_1 * x_i + b_1). \quad (6)$$

B. Local Feature Extraction Using DLPP

The target of DLPP is to find a set of projection vectors $V \in \mathbf{R}^{z \times a}$. After feature extraction by SAE, the data become as $X_{\text{SAE}} = \{x_{\text{SAE}1}, x_{\text{SAE}2}, \dots, x_{\text{SAE}n}\}$, where each sample is a k -dimensional vector. This section uses DLPP for feature extraction from X_{SAE} . The objective function of DLPP is shown in the following equation:

$$v = \arg \min_v \frac{\sum_{k=1}^K \sum_{i,j=1}^{n_k} (v^T x_{\text{SAE}i}^k - v^T x_{\text{SAE}j}^k)^2 Q_{ij}^k}{\sum_{i,j=1}^K (m_i - m_j)^2 T_{ij}} \quad (7)$$

where $m_i = (1/n_i) \sum_{k=1}^{n_i} x_i^k$ and $m_j = (1/n_j) \sum_{k=1}^{n_j} x_j^k$ are the means of the i th and the j th class samples. Q_{ij}^k is the weight relationship between any two samples in the class K and the expression of Q_{ij}^k is shown in the following equation:

$$Q_{ij}^k = \exp\left(-\|x_i^k - x_j^k\|^2 / \varsigma\right). \quad (8)$$

In (7), T_{ij} represents the weight relationship between the mean vector of the sample in class i and the sample in class j . The expression of T_{ij} is shown in the following equation:

$$T_{ij} = \exp(-\|m_i - m_j\|^2/\xi). \quad (9)$$

The denominator of (7) can be transformed to (10) by a simple algebraic derivation

$$\begin{aligned} & \frac{1}{2} \sum_{i,j=1}^K (m_i - m_j)^2 T_{ij} \\ &= \frac{1}{2} \sum_{i,j=1}^K \left(\frac{1}{n_i} \sum_{k=1}^{n_i} y_k^i - \frac{1}{n_j} \sum_{k=1}^{n_j} y_k^j \right)^2 T_{ij} \\ &= \frac{1}{2} \sum_{i,j=1}^K \left(\frac{1}{n_i} \sum_{k=1}^{n_i} v^T x_k^{\text{SAE}i} - \frac{1}{n_j} \sum_{k=1}^{n_j} v^T x_k^{\text{SAE}j} \right)^2 T_{ij} \\ &= \frac{1}{2} \sum_{i,j=1}^K \left[v^T \left(\frac{1}{n_i} \sum_{k=1}^{n_i} x_k^{\text{SAE}i} \right) - v^T \left(\frac{1}{n_j} \sum_{k=1}^{n_j} x_k^{\text{SAE}j} \right) \right]^2 T_{ij} \\ &= \frac{1}{2} \sum_{i,j=1}^K [v^T m_i - v^T m_j]^2 T_{ij} \\ &= \sum_{i=1}^K v^T m_i F_{ii} m_i^T v - \sum_{i,j=1}^K v^T m_j T_{ij} m_j^T v \\ &= v^T M(F - T)M^T v = v^T \mathbf{MSM} v. \end{aligned} \quad (10)$$

Similarly, the molecule of (7) can be transformed to the following equation:

$$\begin{aligned} & \frac{1}{2} \sum_{k=1}^K \sum_{i,j=1}^{n_k} (y_i^k - y_j^k)^2 Q_{ij}^k \\ &= \frac{1}{2} \sum_{k=1}^K \sum_{i,j=1}^{n_k} (v^T x_i^{\text{SAE}k} - v^T x_j^{\text{SAE}k})^2 Q_{ij}^k \\ &= \sum_{k=1}^K \left(\sum_{i,j=1}^{n_k} v^T x_i^{\text{SAE}k} D_{ii}^k (x_i^{\text{SAE}k})^T v - \sum_{i,j=1}^{n_k} v^T x_i^{\text{SAE}k} Q_{ij}^k (x_j^{\text{SAE}k})^T v \right) \\ &= \sum_{k=1}^K v^T X_k (D_k - Q_k) X_k^T v \\ &= v^T X_{\text{SAE}} (D - Q) X_{\text{SAE}}^T v = v^T X_{\text{SAE}} U X_{\text{SAE}}^T v \end{aligned} \quad (11)$$

where Q_k means the similarity weight matrix of the k th class, T represents the weight relationship between the mean vector of samples. D_k and F are diagonal arrays; M can be represented as $M = [m_1, m_2, \dots, m_K] \in \mathbf{R}^{z \times K}$; the corresponding equation takes the form as follows:

$$\begin{aligned} F_{ii} &= \sum_j T_{ij} \\ D_{ii}^k &= \sum_i Q_{ij}^k, \quad D = \begin{bmatrix} D_1 & 0 & 0 \\ 0 & D_2 & 0 \\ 0 & 0 & D_K \end{bmatrix}, \quad Q = \begin{bmatrix} Q_1 & 0 & 0 \\ 0 & Q_2 & 0 \\ 0 & 0 & Q_K \end{bmatrix}. \end{aligned} \quad (12)$$

$$(13)$$

Then, the transformed molecule in (11) and the transformed denominator in (10) are brought into the original objective function (7). The objective function of (7) is transformed into the following equation:

$$v = \arg \min J(v) = \arg \min_v \frac{v^T X_{\text{SAE}} U X_{\text{SAE}}^T v}{v^T \mathbf{MSM}^T v}. \quad (14)$$

The final solution of (14) can be equivalent to the solution of the following equation:

$$X_{\text{SAE}} U X_{\text{SAE}}^T v_a = \lambda_a \mathbf{MSM}^T v_a \quad (15)$$

where $X_{\text{SAE}} U X_{\text{SAE}}^T$ is a $z \times z$ matrix and the matrix $\mathbf{MSM}^T$ has the rank of $R \leq K$. The conventional DLPP is equivalent to the eigenvalue decomposition of the matrix $(\mathbf{MSM}^T)^{-1} (X_{\text{SAE}} U X_{\text{SAE}}^T)$. In this article, SVD is used to decompose the eigenvalues of the matrix $(\mathbf{MSM}^T)^{-1} (X_{\text{SAE}} U X_{\text{SAE}}^T)$ to solve the problem of eigenvalue decomposition due to the dissatisfaction rank of the matrix $\mathbf{MSM}^T$. Moreover, the eigenvalues obtained by SVD have good representativeness and finally the corresponding eigenvectors can be obtained according to the magnitude of the eigenvalues. In the end, the projection matrix $V = \{v_1, v_2, \dots, v_a\}$ is obtained. The final DR data obtained by DLPP is shown in (16); the final data after DR is represented as X_{new}

$$x_{\text{new}i} = x_{\text{SAE}i} \times V. \quad (16)$$

C. Akaike Information Criterion

The DR order is an important parameter. The fault diagnosis performance is related to the DR order. The AIC is used to find the optimal DR order. The expression of AIC is listed in the following equation:

$$a' = \arg \min \text{AIC} = \arg \min f(a) + \frac{a}{n} \quad (17)$$

where $f(a)$ is the model prediction error term representing the misspecification rate of the training data for the first a vectors obtained by projecting the data; n is the average of the various types of observations. The training set misclassification rate $f(a)$ also reflects the amount of useful information about the classification of the first a vectors obtained by DR. Obviously, the misclassification rate decreases as a increases; however, for the test data, the misclassification rate first decreases and increases later. Thus, the model complexity term can effectively suppress the dimensionality increase.

D. Fault Diagnosis Using AdaBoost

After DR of data, the AdaBoost classifier is adopted to identify the fault type for developing fault diagnosis. The decision tree classifier is used as the weak classifier in AdaBoost. The detailed steps of the AdaBoost algorithm are listed as follows.

1) Determine the training data

$$\begin{aligned} S &= \{(x_{\text{new}1}, y_1), (x_{\text{new}2}, y_2), \dots, (x_{\text{new}n}, y_n)\} \\ x_{\text{new}i} &\in X_{\text{new}} (i = 1, 2, \dots, n), \quad y_i \in Y = 1, 2, \dots, K \end{aligned} \quad (18)$$

where $\mathbf{x}_{newi}$ is the data obtained by (16) after DR by SAE and DLPP and $y_i \in Y$ is the category label.

2) Initialize weights

$$\mathbf{C}_1 = (\omega_{11}, \omega_{12}, \dots, \omega_{li}, \dots, \omega_{ln}), \quad \omega_{li} = \frac{1}{n}, \quad i = 1, 2, \dots, n. \quad (19)$$

3) Suppose the number of iterations is $t = 1, 2, \dots, T$ and each learning yields a basic classifier. Repeat the following operation T times.

4) The basic classifier $G_t(x)$ is learned using the training dataset with the current distribution D_t .

5) Obtain the classification error rate e_t . $\mathbf{x}_{newi}$ of the basic classifier $G_t(x)$ on the training data represents the i th class of training data

$$e_t = \sum_{i=1}^K P(G_t(\mathbf{x}_{newi}) \neq y_i) = \sum_{i=1}^K \omega_{ti} I(G_t(\mathbf{x}_{newi}) \neq y_i). \quad (20)$$

a) Calculate the coefficient of $G_t(x)$

$$\alpha_t = \frac{1}{2} \ln \frac{1 - e_t}{e_t}. \quad (21)$$

b) Update the training data weight distribution in preparation for the next iteration.

$$\omega_{t+1,i} = \begin{cases} \frac{\omega_{ti}}{Z_t} e^{-\alpha_t}, & G_t(\mathbf{x}_{newi}) = y_i \\ \frac{\omega_{ti}}{Z_t} e^{\alpha_t}, & G_t(\mathbf{x}_{newi}) \neq y_i \end{cases} \quad (22)$$

$$Z_t = \sum_{i=1}^K \omega_{ti} \exp(-\alpha_t y_i G_m(\mathbf{x}_i)) \quad (23)$$

where Z_t is the normalization factor.

6) Linearly combine the basic classifiers

$$f(x) = \sum_{t=1}^T \alpha_t G_t(x). \quad (24)$$

7) Finally, the strong classifier is obtained

$$G(x) = \text{sign}(f(x)) = \text{sign}\left(\sum_{t=1}^T \alpha_t G_t(x)\right). \quad (25)$$

IV. CASE STUDIES

The performance of SAEDLPP-based fault diagnosis is verified by comparing SAEDLPP-based fault diagnosis with other methods through simulation applications to TEP.

A. Tennessee Eastman Process

The TEP flowchart is shown in Fig. 2. TEP [28] is a realistic simulation platform developed by the Eastman Chemical Company based on actual chemical reaction processes. The data generated from TEP are nonlinear, dynamic, and strongly coupled. TEP can be divided into five parts: a condenser, a reactor, a gas-liquid separator, a compressor, and a stripper. Fifty-two variables are included in TEP, including 41 measured

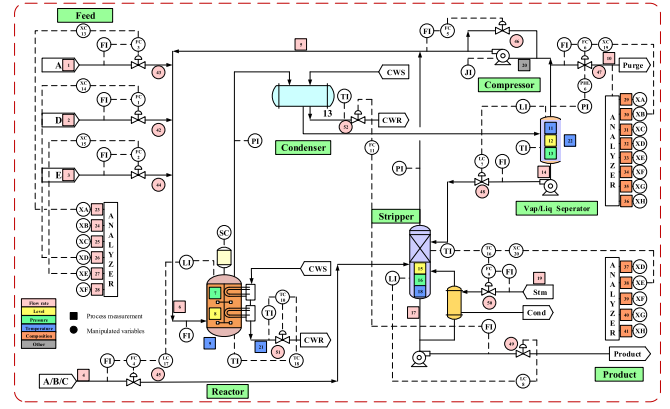


Fig. 2. Flowchart of TEP.

TABLE I
FAULT TYPES OF THE SELECTED SIX FAULTS

Simulation Set	Fault Type
IDV4	Step
IDV5	Step
IDV7	Step
IDV10	Random variation
IDV12	Random variation
IDV14	Sticking

variables and 11 control variables [28]. There are 21 fault types in TEP; the first 16 of these faults are known faults and the last 5 are unknown faults. The sampling time is 3 min in our study. In simulations, 480 training samples and 800 testing samples are generated, respectively.

B. Multifault Case

To verify the superior performance of SAEDLPP-based fault diagnosis, six faults including fault 4, fault 5, fault 7, fault 10, fault 12, and fault 14 are selected as a set of experimental data. The fault types of the selected six faults are shown in Table I. The simulation results are compared with PCA, LPP, and DLPP.

C. Simulation Results and Analysis

Each fault data contain 480 training data and 800 testing data. Determining the DR order is one of the important steps in the experiment. Different dimension orders have different impacts on the final classification results. To determine the DR order, the AIC information criterion is adopted. As shown in Fig. 3, the blue “+” represents the optimal DR order of SAEDLPP found by AIC. As shown in Fig. 3, the optimal DR order is 18 for SAEDLPP, where the algorithm achieves the lowest misclassification rate. In the same way, the optimal DR order of PCA, LPP, and DLPP can be also determined. In Fig. 4, based on the AdaBoost classifier, the accuracy rates of the four methods on the training data are clearly shown. It can be seen that the accuracy rate of the SAEDLPP algorithm proposed in this article is much higher than the other three methods.

In Fig. 3, it can be noticed that the DR order of SAEDLPP is determined as 18 according to the AIC information criterion;

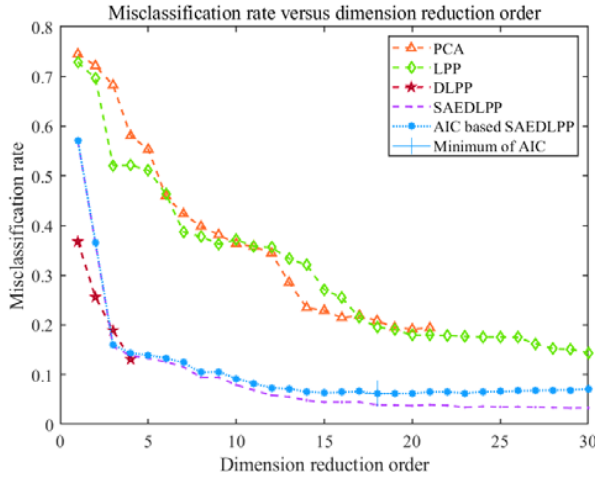


Fig. 3. Misclassification rate of different methods with different orders of reduced dimension.

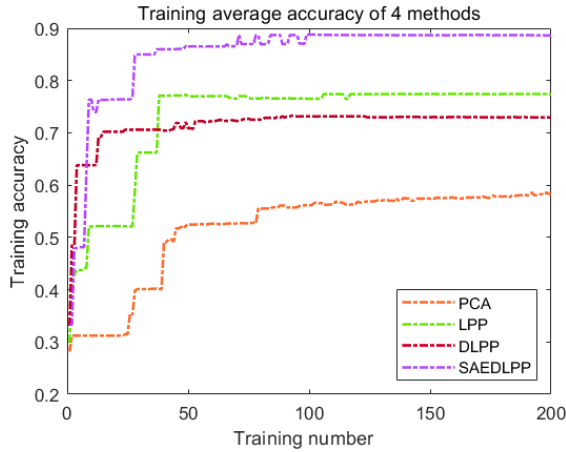


Fig. 4. Accuracy of four different methods on training data.

TABLE II

COMPARISON RESULTS WITH FOUR DIFFERENT METHODS

Fault	PCA (%)	LPP (%)	DLPP (%)	SAEDLPP (%)
Faults 4	71.88	82.75	93.88	90.00
Faults 5	63.63	87.63	95.63	92.75
Faults 7	86.88	81.75	97.00	100
Faults 10	2.75	84.25	2.50	84.25
Faults 12	54.50	38.25	37.13	76.63
Faults 14	59.75	85.75	85.63	82.63
Average accuracy	56.56	76.73	68.62	87.71

it can be seen that the maximum DR order of DLPP is not more than the number of fault classes c due to the problem of matrix decomposition. The DR order of DLPP is determined as four in this article. The diagnosis results of the four fault diagnosis models on the testing data are listed in Table II. As shown in Table II, the average accuracy of the SAEDLPP algorithm proposed in this article is much higher than those of the other three algorithms of PCA, LPP, and DLPP. In the testing dataset, the average accuracy of PCA, LPP, and DLPP are 56.56%, 76.73%, and 68.62%, respectively; while the average accuracy of the SAEDLPP algorithm proposed in this article reaches 87.71%; especially for fault 12, the accuracy of the presented SAEDLPP

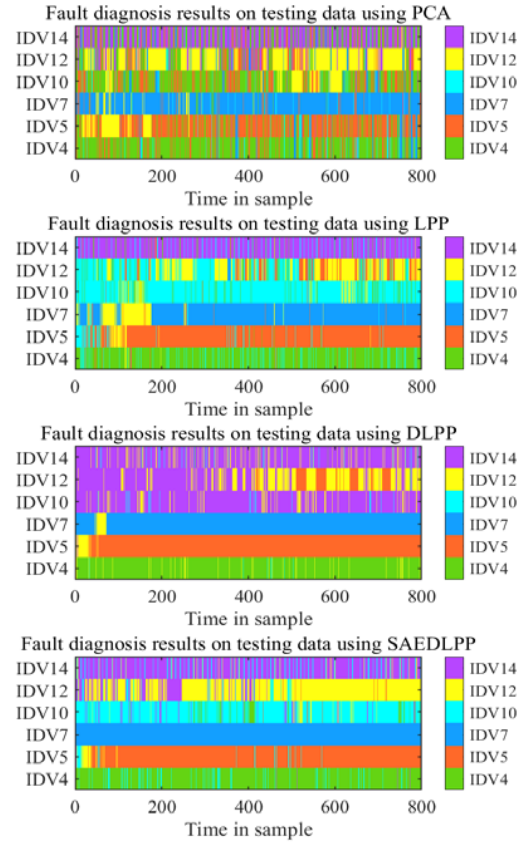


Fig. 5. Results of the fault diagnosis performed on the test data using four methods.

algorithm is significantly improved compared with the other three algorithms and the diagnosis rate of fault 7 by SAEDLPP reaches 100%, indicating that the proposed SAEDLPP-based fault diagnosis can achieve acceptable diagnosis accuracy.

Fig. 5 shows the diagnosis rate of each type of fault. The right side of Fig. 5 shows the corresponding color when each type of fault is correctly diagnosed; thus, the more consistent the color of each type of fault, the higher the diagnosis rate of the corresponding fault. It can be clearly seen from Fig. 5, the proposed SAEDLPP obtains more consistent colors than other methods, which also indicates that SAEDLPP achieves better fault diagnosis accuracy than the other three methods.

D. Visualization Analysis

The first three dimensions of the extracted features from the four algorithms of PCA, LPP, DLPP, and SAEDLPP after DR are demonstrated. The visualization results of the first three dimensions of the extracted features from the four algorithms of PCA, LPP, DLPP, and SAEDLPP on the testing dataset are shown in Fig. 6. Fig. 6 shows the visualization results of the testing dataset of the six faults of fault 4, fault 5, fault 7, fault 10, fault 12, and fault 14. From Fig. 6, the extracted features of the SAEDLPP algorithm presented in this article can be well discriminated from the six faults when mapping the high-dimensional data to the 3-D space and so SAEDLPP-based fault diagnosis can obviously achieve higher

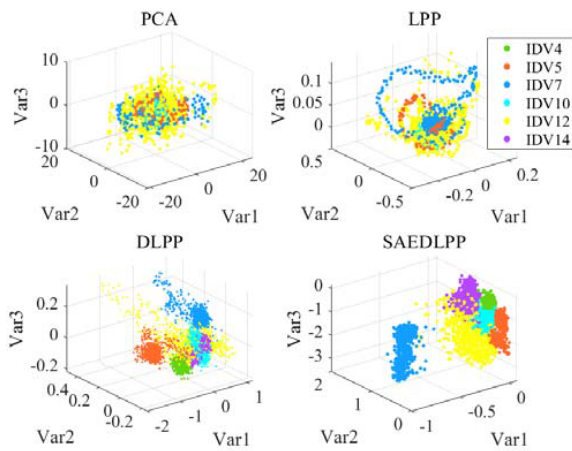


Fig. 6. Visualization of the first three extracted features using four methods.

classification accuracy than the other three algorithms of PCA, LPP, and DLPP. Through visualization, it can be proved that the presented SAEDLPP-based fault diagnosis can achieve higher classification accuracy than PCA, LPP, and DLPP.

V. CONCLUSION

In order to effectively enhance the performance of fault diagnosis in industrial processes, this article proposes a fault diagnosis methodology based on SAEDLPP. First, SAE is used to extract features from the original data, and then the newly obtained data after feature extraction are used to obtain the projection matrix through the manifold learning algorithm of DLPP, which can realize the DR of high-dimensional data by reasonably combining the advantages of two feature extraction methods; in order to solve the matrix decomposition problem in SAEDLPP, SVD is used; finally, an AdaBoost classifier is used to classify the data after DR to obtain fault types. To verify and confirm the superior effectiveness of SAEDLPP, different types of faults in TEP are used for experiments. Simulation results show that the presented SAEDLPP can achieve better feature extraction capability and much higher fault diagnosis accuracy compared with PCA, LPP, and DLPP.

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Yan-Lin He (Member, IEEE) received the B.Sc. and Ph.D. degrees from the College of Information Science and Technology (CIST), Beijing University of Chemical Technology (BUCT), Beijing, China, in 2011 and 2016, respectively.

He is currently a Professor with CIST, BUCT. His research interests include computational intelligence, artificial intelligence, machine learning, data mining, fault diagnosis, and process modeling.



Yuan Xu received the B.Sc. and Ph.D. degrees from the College of Information Science and Technology (CIST), Beijing University of Chemical Technology (BUCT), Beijing, China, in 2005 and 2010, respectively.

She is currently a Professor with CIST, BUCT. Her research interests include computational intelligence, fault diagnosis, data mining, and process modeling.



Kun Li received the bachelor's degree in engineering from the Mechanical College of Jinzhong University, Jinzhong, China, in 2016. He is currently pursuing the master's degree with the College of Information Science and Technology (CIST), Beijing University of Chemical Technology (BUCT), Beijing, China.

His research interests include computational intelligence, machine learning, data mining, and fault diagnosis.



Ning Zhang is currently pursuing the master's degree with the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing, China. His major is control science and engineering.

His research interests are fault diagnosis, small sample size modeling, and machine learning.



Qun-Xiong Zhu received the Ph.D. degree from the College of Information Science and Technology (CIST), Beijing University of Chemical Technology (BUCT), Beijing, China, in 1996.

He is currently a Professor with CIST, BUCT. His research interests include computational intelligence, artificial intelligence, machine learning, data mining, fault diagnosis, and process modeling.